

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I GK Vyshnavi / 1924²⁵⁰⁶⁹ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

19/20
25069

Subnetting for 192.168.20.0/24

given

network: 192.168.20.0/24

lab A $\rightarrow$ /26

lab B $\rightarrow$ /27

lab C $\rightarrow$ /28

Formula:

Total addresses = $2^{(32 - \text{prefix})}$

usable Hosts = Total addresses - 2

lab A (/26):

Subnet mask:

255.255.255.192

Total addresses: $2^{32-26} = 64$

usable Hosts: $64 - 2 = 62$

Network address - 192.168.20.0

First host - 192.168.20.1

last host - 192.168.20.62

Broadcast address - 192.168.20.63

usable Hosts - 62.

lab B (/27):

Next available subnet begins at 192.168.20.64

Subnet mask:

Total addresses: $2^5 = 32$

usable Hosts: $32 - 2 = 30$

network address - 192.168.20.64

First Host - 192.168.20.65

last Host - 192.168.20.94

Broad cast address - 192.168.20.95

usable Hosts - 30.

lab C (/28):

Next Subnet begins at 192.168.20.96

Subnet mask:

255.255.255.240

Total address :- $2^4 = 16$

usable Hosts: $16 - 2 = 14$

Network address - 192.168.20.96

First Host - 192.168.20.97

last Host - 192.168.20.110

Broad Cast - 192.168.20.111

usable Hosts - 14.

2. Systems with mask 255.255.255.192 (/26)

Given.

Subnet mask = 255.255.255.192

Block size - $256 - 192 = 64$

Subnet

1. 0-63

2. 64-127

3. 128-191

4. 192-255

System 1

IP: 192.168.1.130

Falls inside

128-191

network address - 192.168.1.128

First host - 192.168.1.129

Last host - 192.168.1.190

Broadcast - 192.168.1.191

IP 192.168.1.130 is Valid

System 2

IP: 192.168.1.190

also falls in

128-191

network address - 192.168.1.128

First Host - 192.168.1.129

Last Host - 192.168.1.190

Broadcast - 192.168.1.191

IP: 192.168.1.190 is Valid

usable Host: $64 - 2 = 62$

3. organization Network 172.16.0.0/16

Given

network:

172.16.0.0/16

Departments:

HR $\rightarrow /25$

Sales $\rightarrow /26$

IT $\rightarrow /27$

admin $\rightarrow /28$

HR (125)

network - 172.16.0.0

Host Range - 172.16.0.1 - 172.16.0.126

Broadcast - 172.16.0.127

usable Hosts - 126

sales (126)

starts at 172.16.0.128

network - 172.16.0.128

Host range - 172.16.0.129 - 172.16.0.190

Broadcast - 172.16.0.191

usable Hosts - 62

It (127)

start at 172.16.0.192

network - 172.16.0.192

Host Range - 172.16.0.193 - 172.16.0.222

Broadcast - 172.16.0.223

usable Hosts - 30

Admin (128)

starts at 172.16.0.224

network - 172.16.0.224

Host range - 172.16.0.225 - 172.16.0.238

Broadcast - 172.16.0.239

usable Host - 14.

Network

172.16.10.0/24 using /27

step 1: no. of subnets.

Borrowed Bits: $27 - 24 = 3$

subnets - $2^3 = 8$

step 2: Hosts:

Host Bits: $32 - 27 = 5$

address: $2^5 = 32$

usable: $32 - 2 = 30$

step 3: subnets

sub net	Range
0	0 - 31
1	32 - 63
2	64 - 95
3	96 - 127
4	128 - 159
5	160 - 191
6	192 - 223
7	224 - 255

step 4: IP 172.16.10.78

falls inside

64 - 95

network addresses:

172.16.10.64

Broadcast:

172.16.10.95